Part I used three symmetry elements as a yes/no test. Part II states the criterion properly, because the harder JEE Advanced items — and every isomer-counting question with a ring — need the full set.

## 6.1 The five operations

| Element | Operation | Symbol | Improper? |
|---|---|---|---|
| identity | do nothing | $E$ | — |
| proper axis | rotate 360/n | $C_n$ | no |
| plane | reflect | $\sigma$ | yes |
| centre | invert through a point | $i$ | yes |
| alternating axis | rotate 360/n, then reflect in the plane perpendicular to the axis | $S_n$ | yes (n even), no (n odd) |

Table 6.1 — the complete inventory for molecular symmetry.

Only the **improper** operations ($\sigma$, $i$, $S_n$) change the handedness of an object. A molecule is chiral if and only if it has no improper operation. Proper rotations $C_n$ preserve handedness, so a molecule with a $C_3$ axis can be perfectly chiral.

> KEY RESULT
> Chiral ⇔ no $\sigma$, no $i$, no $S_n$. Not "no symmetry": a chiral molecule may have as many proper axes as it likes.

## 6.2 The identities worth knowing

- **$S_1 = \sigma$.** Rotate by 360° (do nothing) and reflect: that is a plain reflection in a plane.
- **$S_2 = i$.** Rotate by 180° and reflect in the perpendicular plane: that is inversion.
- **$S_n$ with n even implies a $C_{n/2}$ axis**, so $S_4$ implies $C_2$, $S_6$ implies $C_3$. Improper axes therefore do not add new chirality decisions beyond $\sigma$ and $i$ for small $n$.
- **$\sigma$ and $i$ together imply a $C_2$ axis** through their intersection, which is why trans-decalin (i) also has a $C_2$ .

## 6.3 Worked symmetry assignments

| Molecule | Elements present | Chiral? |
|---|---|---|
| $H_2O$ | $C_2$, two $\sigma_v$ | no |
| $NH_3$ | $C_3$, three $\sigma_v$ | no |
| $CH_4$ | 4 $C_3$, 3 $S_4$, 6 $\sigma_d$ | no |
| CHFClBr | none ($C_1$) | yes |
| staggered ethane | $C_3$, 3 $C_2$, $i$, $S_6$ | no |
| chair cyclohexane | $C_3$, 3 $C_2$, 3 $\sigma_d$, $i$ | no |
| twist-boat cyclohexane | $C_2$, no $\sigma$, no $i$, $S_n$ | yes (a pair) |

Table 6.2 — assignments to check your intuition against.

## 6.4 Asymmetric versus dissymmetric

A molecule with no element at all except the identity (point group $C_1$) is **asymmetric** and also chiral. A molecule with proper axes but no improper element (point group $D_n$, for example) is **dissymmetric**: it is chiral, but it has rotational symmetry. This is the precise version of the statement "all asymmetric molecules are chiral, but not all chiral molecules are asymmetric".

## 6.5 Why the test is enough

If an improper operation maps the molecule onto itself, there is a rigid motion that carries the mirror image onto the original — so the two are superimposable and the molecule is achiral. Conversely, if the mirror image can be superimposed on the original, the operation that does it is improper. That is the proof in one line, and it is worth writing in an answer.

## 6.6 Symmetry of the conformers of ethane

Staggered ethane ($D_{3d}$) has a $C_3$ axis, three $C_2$ axes perpendicular to it, a centre of symmetry, an $S_6$ axis and three $\sigma_d$ planes. Eclipsed ethane ($D_{3h}$) has a $C_3$ axis, three $C_2$ axes, a $\sigma_h$ plane and three $\sigma_v$ planes, and no centre. Both are achiral, but for different reasons.

| Species | Proper axes | Improper elements | Chiral? |
|---|---|---|---|
| staggered ethane | $C_3$, 3 $C_2$ | $i$, $S_6$ (and $\sigma_d$) | no |
| eclipsed ethane | $C_3$, 3 $C_2$ | $\sigma_h$, $S_3$ | no |
| gauche butane | $C_2$ | none | yes (conformational) |
| anti butane | $C_{2h}$ | $i$, $\sigma_h$ | no |
| chair cyclohexane | $C_3$, 3 $C_2$ | $i$, $S_6$, $\sigma_d$ | no |
| twist-boat cyclohexane | $C_2$ | none | yes (a pair) |

Table 6.3 — the same molecule can be chiral in one conformation and achiral in another.

The twist-boat of cyclohexane is chiral, and the two twist-boats are enantiomers. They interconvert through the (achiral) chair at a rate of about $10^5\ s^{-1}$ at room temperature, so the population of the chiral species is negligible, and cyclohexane itself is achiral. At low temperature the interconversion slows and the chirality of a fleeting species can be observed spectroscopically.

## 6.7 A final checklist for "is it chiral?"

1. Draw the most symmetric conformation you can find. If it has $\sigma$ or $i$, the molecule is achiral — stop.
2. If no improper element is present, look for a stereocentre, a chiral axis or a chiral plane to explain the chirality.
3. If the molecule is flexible, ask whether the conformers interconvert at room temperature. If they do, the sample is achiral even if individual conformers are chiral.
4. If the interconversion is slow (atropisomerism, a rigid ring), the conformer is the molecule, and the chirality is observable.

TRAP
Point-group labels are a convenience, not a requirement. If you can point to a plane, a centre or an $S_n$ axis, you have already decided the question; naming the group adds nothing to the chemistry.

## 6.8 The chirality axis

In an allene the four substituents lie in two perpendicular planes, and the molecule has a chiral axis along the C=C=C line. If both ends carry two different groups, the molecule is chiral. The two enantiomers are labelled aR and aS (or P and M); the priority rule for the axis is the same CIP apparatus applied to each end, and the descriptors follow from the arrangement of the higher-priority groups.

For JEE it is enough to know that the enantiomers exist and are not labelled with R/S at a carbon. If you do assign the axial descriptor, look down the axis, rank the two groups on each end, and compare the near and far ends exactly as you would for a stereocentre.

## 6.9 Cumulenes with more than two double bonds

| Number of C=C | End planes | Chirality (both ends unsymmetrical) | Geometrical isomerism |
|---|---|---|---|
| 2 | perpendicular | yes, chiral axis | no |
| 3 | coplanar | no | yes, E/Z |
| 4 | perpendicular | yes, chiral axis | no |
| 5 | coplanar | no | yes, E/Z |

Table 6.4 — the odd/even rule continued indefinitely; each additional double bond rotates the far plane by 90°.

The reason is that each additional double bond rotates the plane of the far end by 90°: with an even number of double bonds the planes are perpendicular (no common reference plane, hence chirality), and with an odd number they are coplanar (a common reference plane, hence cis/trans).

## 6.10 Spiro compounds

Two rings sharing a tetrahedral carbon lie in perpendicular planes. If each ring carries different substituents on its two sides, the molecule has a chirality axis through the spiro carbon. The decisive test is still a mirror plane: if either ring is symmetric about the plane containing the other ring, the molecule is achiral.

KEY RESULT
Draw each ring separately and ask whether it has a plane through the spiro carbon. Two unsymmetrical rings → chiral axis; one symmetrical ring → achiral.

## 6.11 Biphenyls and atropisomerism

The inter-ring bond of a biphenyl is a single bond, but rotation is hindered by the ortho hydrogens. With four bulky ortho substituents the barrier is 80–160 kJ mol$^{-1}$, high enough to isolate the two enantiomers at room temperature. This is atropisomerism — chirality from restricted rotation about a single bond.

| Ortho substitution | Barrier | Isolable at 25 °C? |
|---|---|---|
| none (biphenyl) | ≈ 8 kJ mol$^{-1}$ | no |
| 2,2′-dimethyl | ≈ 80 kJ mol$^{-1}$ | borderline |
| 2,2′-di-tert-butyl | ≈ 150 kJ mol$^{-1}$ | yes |
| 2,2′,6,6′-tetramethyl | ≈ 150 kJ mol$^{-1}$ | yes |

Table 6.5 — atropisomerism needs bulk, not just a substituent.

The classic example is 2,2′,6,6′-tetramethylbiphenyl-3,3′-dicarboxylic acid (the "Mills-Nixon" molecule, whose two enantiomers were resolved in 1933 — the first proof that restricted rotation alone can generate chirality). It sits in the same chapter because the argument is the same one used for allenes: a locked axis, two unsymmetrical ends, no plane.

TRAP
Ortho substituents alone are not enough: if each ring is symmetrically substituted (for example 2,2′,6,6′-tetramethylbiphenyl with identical groups at 3 and 5), a plane exists and the compound is achiral. Both conditions are needed.

## 6.11b Naming and drawing an atropisomer

To draw the two enantiomers of a 2,2′,6,6′-tetrasubstituted biphenyl, draw the two rings in perspective, then hold one ring fixed and reflect the other through the plane of the first. The two drawings are mirror images and cannot be superimposed by rotation about the inter-ring bond, because that rotation is blocked. A common exam task is to decide whether a given substitution pattern is chiral; the four-question test below settles it without drawing both rings.

1. Are all four ortho positions substituted? If not, rotation is free and the molecule is achiral.
2. Are the two rings identically substituted? If yes, a plane perpendicular to the axis survives and the molecule is achiral.
3. Is there a $C_2$ axis along the inter-ring bond with no plane? Then the molecule is chiral.
4. Does the substitution give a plane through both rings (for example identical groups at 3 and 5 on one ring)? Then the molecule is achiral.

| Pattern | Chiral? | Deciding feature |
|---|---|---|
| 2,2′-dimethylbiphenyl | no | two identical halves; free enough to rotate |
| 2,2′-dimethyl-6,6′-dinitrobiphenyl | yes | four different ortho groups, locked |
| 2,2′,6,6′-tetramethylbiphenyl | no | plane perpendicular to the axis |
| 2,2′,6,6′-tetramethylbiphenyl-3,3′-dicarboxylic acid | yes | the 3 and 3′ groups break the plane |
| BINOL | yes | no plane; the two naphthyl halves are locked |

Table 6.6 — atropisomerism needs both bulk and asymmetry.

KEY RESULT
Atropisomerism = restricted rotation + no plane. Bulk supplies the restriction; unsymmetrical substitution supplies the missing plane. Both are needed, and either one missing makes the compound achiral.

## 6.12 Trans-cycloalkenes

A trans double bond inside a ring forces the two allylic carbons onto opposite faces. For rings of seven or fewer atoms this is impossible without absurd strain; cyclooctene is the first that can twist enough. The strain energy of trans-cyclooctene is about 50 kJ mol$^{-1}$ above the cis isomer, and the trans form is chiral because the twist removes every mirror plane.

| Ring | cis | trans | Note |
|---|---|---|---|
| cyclopropene | only | no | trans would need a 60° twist |
| cyclobutene | only | no | highly strained |
| cyclopentene | only | no | half-chair, cis only |
| cyclohexene | only | no | half-chair, cis only |
| cycloheptene | only | no | trans is calculable but not isolable |
| cyclooctene | yes | yes | trans is chiral and isolable |

Table 6.7 — the ring size decides whether the trans alkene exists.

## 6.13 Fused ring junctions

In a fused bicyclic system each junction carbon carries one hydrogen. Cis means both hydrogens on the same face; trans means opposite faces. For decalin both are possible: cis-decalin is flexible and less stable, trans-decalin is rigid and more stable. But here achiral — cis has a plane (on one ring conformation) and trans a centre.

In the steroids the same junction appears at C5/C10 and at C8/C9. The C5, C8, C9 and C14 series differ only in configuration at C5, which is a ring junction; the two series have different shapes, different biological activities, and are diastereomers. This is the most important real-world consequence of ring-junction stereochemistry.

## 6.14 Counting with coupled centres

In a rigid bicyclic system the two bridgehead carbons cannot invert independently. Flipping one would require breaking the bridge, so the pair behaves as **one** stereocentre when you apply $2^n$.

1. Count each coupled bridgehead pair as **one** centre.
2. If the bridgeheads are equivalent by symmetry, the molecule has fewer stereoisomers than the naive count; if they are not, the cis and trans are diastereomers.
3. Do not push the shortcut into a rigid cage. For 2,5-disubstituted bicyclo[2.2.1]heptane the two bridgeheads are coupled, but the C2 and C5 substituents can only point along the two available directions, so some of the $2^n$ drawings close on themselves. Draw the cage, mark each substituent up or down, and discard drawings that superimpose — never quote a number from n alone.

KEY RESULT
The "bridgeheads count as one" shortcut applies to coupled centres only. In a flexible ring such as cyclohexane the two substituted carbons are independent, and each contributes its own factor.

## 6.15 Medium rings

Cyclooctane adopts the boat-chair (crown) conformation; cyclononane has several low-energy conformers; cyclodecane is the smallest ring with a stable transannular interaction. These are not in the JEE syllabus, but they explain why the odd/even rule for trans-alkenes changes at eight: the ring has to be large enough that a trans arrangement is not prohibitively strained.

## 6.16 From energies to populations

The Boltzmann distribution converts an energy difference into a population ratio: $N_i/N_j = \exp(-\Delta E/RT)$. At 298 K, $RT$ = 2.48 kJ mol$^{-1}$, so each 5.7 kJ mol$^{-1}$ of energy costs a factor of 10 in population. This is the single most useful numerical fact in conformational analysis.

## 6.17 A-values

The A-value of a substituent is the free-energy cost of putting it axial in cyclohexane, measured from the equilibrium ratio of the two chairs. It is the quantitative version of the 1,3-diaxial argument.

| Substituent | A-value / kJ mol$^{-1}$ | ≈ % equatorial at 25 °C |
|---|---|---|
| H | 0 | 50 |
| F | 1.0 | 60 |
| $CH_3$ | 7.3 | 95 |
| $CH_2CH_3$ | 7.5 | 95 |
| $CH(CH_3)_2$ | 9.0 | 97 |
| $C(CH_3)_3$ | 20 | >99.9 |
| OH | 4.0 | 84 |
| $OCH_3$ | 2.5 | 73 |

Table 6.8 — A-values; use them to compare two chairs, not to predict a third decimal place.

Note that the A-value of OH is much smaller than that of $CH_3$, even though OH is heavier. The controlling factor is not mass but the length and shape of the bond: a C–O bond is shorter than a C–C bond, so an axial OH reaches less far into the 1,3-diaxial hydrogens.

## 6.18 1,3-diaxial interactions

An axial substituent on C1 points towards the axial hydrogens on C3 and C5. Each such H···X interaction costs roughly 3.5–4 kJ mol$^{-1}$ for a methyl group, and there are two of them, which is where the ≈ 7.3 kJ mol$^{-1}$ A-value comes from. A 1,3-diaxial methyl–methyl interaction (two methyls both axial on C1 and C3) is much larger, which is why 1,3-dimethylcyclohexane strongly prefers the e,e chair.

KEY RESULT
Two 1,3-diaxial interactions ≈ one A-value. That is why the A-value is roughly twice a single H···$CH_3$ interaction, and why 1,3-diaxial methyl–methyl contacts dominate the conformational analysis of substituted cyclohexanes.

## 6.19 The anomeric effect

In a pyranose ring an electronegative substituent at C1 (the anomeric carbon) often prefers the **axial** position, contrary to the steric prediction. The reason is stereoelectronic: an axial C–X σ orbital accepts electron density from a lone pair on the ring oxygen (the n → σ interaction), and that overlap is only possible when X is axial. The effect is worth about 4–8 kJ mol$^{-1}$ in water and larger in non-polar solvents.

This is the first genuinely stereoelectronic effect in the chapter, and it is worth understanding rather than memorising: the same n → σ requirement explains why glycosidic bonds are hydrolysed by an axial-assistance mechanism, and why α-glucose is more stable than the steric argument would predict.

TRAP
The anomeric effect is not a JEE Main topic. In a JEE Advanced answer, one sentence — "the axial C–X bond allows n → σ donation from the ring oxygen" — is enough; do not build an answer around it unless the question mentions anomers.

## 6.20 The ring-flip path

The chair-to-chair interconversion does not pass through the boat. The path is chair → half-chair (transition state, ≈ 45 kJ mol$^{-1}$) → twist-boat (shallow minimum, ≈ 23 kJ mol$^{-1}$) → half-chair → chair. The twist-boat is a genuine intermediate on the path, which is why cooling a cyclohexane sample can, in principle, trap it.

- Chair: all bonds staggered, no strain — the global minimum.
- Twist-boat: one eclipsing interaction removed relative to the boat; ≈ 23 kJ mol$^{-1}$ above the chair.
- Boat: two flagpole hydrogens; ≈ 29 kJ mol$^{-1}$ above the chair; not a minimum on the true path.
- Half-chair: the highest point, ≈ 45 kJ mol$^{-1}$; the C–C–C angles are strained.

## 6.21 Fused rings cannot flip

A fused bicyclic system such as decalin cannot undergo a chair flip without breaking a bond, because the shared bond locks both rings. The consequence is that substituents on a fused system have fixed axial/equatorial assignments, and cis/trans ring junctions are configurationally stable. This is why steroids have a fixed shape and why their stereochemistry is described by the junction configuration rather than by a conformational average.

## 6.22 Bridgehead double bonds (Bredt's rule)

A double bond at a bridgehead of a small bicyclic system would require the bridgehead to be planar, which the cage cannot accommodate. Bredt's rule therefore says that bridgehead double bonds are stable only in large rings (roughly, when the bridge contains at least eight atoms). Norbornene cannot have a bridgehead double bond; the corresponding alkene is isolable only in much larger bicyclic systems.

| System | Bridgehead C=C? | Why |
|---|---|---|
| bicyclo[2.2.1]heptane | no | bridgehead cannot be planar |
| bicyclo[2.2.2]octane | no | same |
| bicyclo[4.4.0]decane (decalin) | yes, at the junction | the ring is large enough |
| bicyclo[3.3.1]nonane | no | still too rigid |

Table 6.9 — Bredt's rule is about the size of the cage, not about the ring count.

KEY RESULT
**Bredt's rule: no double bond at a bridgehead unless the bridge is long enough to let the bridgehead be planar. It is a strain argument, not a rule about atoms.**

TRAP
**Bredt's rule is often quoted as "no double bond at a bridgehead". The exceptions in large rings are real, and JEE Advanced has asked about them; state the rule as a rule of thumb about strain, not as a prohibition.**

## 6.23 Two words examiners test

A reaction is **stereospecific** if a defined stereoisomer of the starting material gives a defined stereoisomer of the product — the stereochemistry is fixed by the mechanism, for every molecule. $S_N2$ is stereospecific: (R) gives (S) and nothing else.

A reaction is **stereoselective** if one stereoisomer is formed preferentially from a substrate that could give several — the stereochemistry is a matter of preference, not of mechanism. Hydrogenation of an alkene bearing a nearby stereocentre is stereoselective: one diastereomer dominates, but both may be formed.

| Reaction | Stereospecific? | Stereoselective? | Why |
|---|---|---|---|
| $S_N2$ | yes | — | backside attack fixes the product |
| E2 | yes | yes (Zaitsev) | anti-periplanar geometry + alkene stability |
| $Br_2$ addition | yes | — | anti addition through the bromonium ion |
| hydrogenation of a substituted alkene | no | yes | the catalyst face is favoured by sterics |
| nucleophilic addition to a chiral aldehyde | no | yes | Cram / Felkin-Anh |

Table 6.10 — the distinction is mechanism versus preference.

KEY RESULT
Stereospecific: every molecule reacts the same way for a geometric reason. Stereoselective: one outcome dominates for an energetic reason. A reaction can be both, but never one without a reason.

## 6.24 Applying the Cram rule

When a nucleophile adds to a carbonyl adjacent to a stereocentre, the Cram rule predicts attack on the less hindered face in the conformation in which the largest group on the stereocentre is eclipsed with the carbonyl oxygen. The Felkin-Anh model refines this with the "large, medium, small" arrangement and is the version used for predicting the major diastereomer in modern answers.

1. Draw the carbonyl with the adjacent stereocentre.
2. Rank the three groups on the stereocentre as large, medium and small.
3. Put the large group perpendicular to the carbonyl plane, with the medium group on the other side.
4. The nucleophile attacks the face opposite the large group — the Bürgi-Dunitz angle is about 107°.

The Felkin-Anh product is the major diastereomer when the stereocentre's groups differ strongly in size; the Cram product often coincides with it for simple alkyl substituents. Both are models — the exam asks you to apply the rule stated in the question, not to decide which model is correct.

## 6.25 Stereochemistry of addition to cis/trans alkenes

This is the table from Part I, restated as a reaction-design principle. For a **cis** alkene, syn addition gives the meso compound and anti addition the racemic pair. For a **trans** alkene it is reversed. So choosing a reagent is choosing a stereochemical outcome: $OsO_4$ (syn) and $Br_2$ (anti) are not interchangeable if you care about the product's symmetry.

| Target | Alkene | Reagent | Outcome |
|---|---|---|---|
| meso diol | cis | $OsO_4$ / $H_2O_2$ (syn) | one compound |
| racemic diol | trans | $OsO_4$ / $H_2O_2$ (syn) | a pair |
| meso dibromide | trans | $Br_2$ (anti) | one compound |
| racemic dibromide | cis | $Br_2$ (anti) | a pair |

Table 6.11 — pick the reagent that gives the symmetry you want.

TRAP
Do not call a reaction "stereospecific" merely because one product predominates. Stereospecificity is a yes/no statement about mechanism; the word for preference is stereoselective.

## 6.25b A worked stereoselective addition

**(R)-2-Methylbutanal + $NaBH_4$.** The aldehyde carbon is prochiral; the adjacent C2 carries $CH_3$, $C_2H_5$, H and CHO. Hydride can attack either face of the carbonyl, giving (2R,3R)- and (2R,3S)-2-methylbutan-1-ol — two diastereomers, not enantiomers, because the existing centre is unchanged.

Which dominates? Apply Felkin-Anh: put the largest group on C2 (ethyl) perpendicular to the carbonyl plane, the medium group (methyl) on the other side, and the small group (H) between them. Hydride attacks the face opposite the large group. The major product is therefore the diastereomer with the new OH anti to the ethyl group — a stereoselective, not stereospecific, reaction.

| Reagent | Facial preference | Reason |
|---|---|---|
| $NaBH_4$, MeOH | Felkin-Anh (opposite the large group) | steric approach control |
| $LiAlH_4$, $Et_2O$ | same, often lower selectivity | smaller effective hydride donor |
| Grignard, $Et_2O$ | Cram chelation possible | if a chelating group is present on C2 |

Table 6.12 — the same substrate, three reagents, three selectivity models.

TRAP
Do not call the two products enantiomers. The existing stereocentre is the same in both, so they are diastereomers — separable by ordinary means, with different physical properties.

Part I counted stereoisomers with two formulas. Those formulas are special cases of one idea — counting orbits under a symmetry group — and once you have the idea you can count cases the formulas do not cover: rings, polyenes, and molecules with several kinds of centre.

## 6.26 Why the formulas fail

The formulas assume a chain whose only symmetry is the exchange of its two ends. A ring has rotational symmetry; a branched chain has a different automorphism group; a molecule with two different kinds of centre has none of the simple structure. The general method — count arrangements, then divide out the symmetry — works in every case.

## 6.27 Burnside's lemma in one line

The number of distinct objects is the average number of arrangements fixed by each symmetry operation:

$$N = (1/|G|)\ \Sigma_g\ |\mathrm{Fix}(g)|$$

For a symmetric chain the group has two elements: the identity, which fixes all $2^n$ arrangements, and the end-exchange, which fixes only the arrangements that read the same from both ends. That gives $N = (2^n + F)/2$, where $F$ is the number of palindromic arrangements. For a chain the exchange reverses the sequence and inverts every descriptor, so $F = 2^{\mathrm{ceil}(n/2)}$.

$$N = (2^n + 2^{\mathrm{ceil}(n/2)}) / 2$$

| $n$ | $2^n$ | $2^{\mathrm{ceil}(n/2)}$ | $N$ | matches Part I? |
|---|---|---|---|---|
| 1 | 2 | 2 | 2 | yes (but-2-ene) |
| 2 | 4 | 2 | 3 | yes (tartaric acid) |
| 3 | 8 | 4 | 6 | yes |
| 4 | 16 | 4 | 10 | yes |
| 5 | 32 | 8 | 20 | yes |
| 6 | 64 | 8 | 36 | yes |

Table 6.13 — Burnside reproduces every row of the Part I table.

KEY RESULT
For an even-$n$ symmetric chain, $N = (2^n + 2^{(n/2)})/2$, identical to the table in Part I. Burnside explains why the formulas have that shape.

## 6.28 Counting stereoisomers of a symmetric molecule

**Tartaric acid ($n = 2$, ends exchanged by a $C_2$-type operation).** The identity fixes all four configurations; the exchange fixes the two strings that read the same from both ends, (R,S) and (S,R) — the anti-palindromes — so the orbit count is $(4 + 2)/2 = 3$ stereoisomers: the active pair (R,R)/(S,L) and the meso compound (R,S) ≡ (S,R).

**A trisubstituted benzene.** Positions 1, 3, 5 with three different substituents: the ring has a $C_3$ axis and three mirror planes. There is no stereocentre and no chiral arrangement, so the count is the number of positional isomers, which is an orbit count of the substitutions under the ring's symmetry — three distinct isomers for three identical substituents on 1,3,5, and more when they differ.

## 6.29 Polysubstituted rings

| Ring | Substitution | Distinct positional isomers | Method |
|---|---|---|---|
| benzene | $C_6H_4Cl_2$ | 3 | ortho, meta, para |
| benzene | $C_6H_3Cl_3$ | 3 | 1,2,3; 1,2,4; 1,3,5 |
| benzene | $C_6H_2Cl_4$ | 3 | 1,2,3,4; 1,2,3,5; 1,2,4,5 |
| cyclopropane | $C_3H_4Cl_2$ | 2 | 1,1 and 1,2 |
| cyclobutane | $C_4H_6Cl_2$ | 3 | 1,1; 1,2; 1,3 |

Table 6.14 — positional isomerism is an orbit count, not a stereochemistry question.

TRAP
Do not apply the stereoisomer formulas to a positional-isomer question. "How many dichlorobenzenes?" is a symmetry count on the ring; "how many stereoisomers of 1,2-dichlorocyclohexane?" is a stereochemistry count. Read which one is asked.

## 6.29b Counting with two different symmetry operations

A molecule may have more than one operation that maps it onto itself, and each contributes its own fixed-point count. The tetramethylcyclobutane case of sheet J-05 is the smallest example: the square has a four-fold rotation and two kinds of mirror plane, and the up/down patterns must be counted under all of them, not just under a single flip.

| Molecule | Symmetry operations | Orbit count | Result |
|---|---|---|---|
| 1,2,3,4-tetramethylcyclobutane | $E$, $C_2$, $C_4$, $C_4^3$, 4 σ | 16 → 3 | 3 achiral stereoisomers |
| benzene, $C_6H_3Cl_3$ | $D_{6h}$ (12 operations) | — | 3 positional isomers |
| tartaric acid | $E$, end-exchange | (4 + 2)/2 | 3 stereoisomers |
| 2,3,4-trihydroxypentanedioic acid | $E$, end-exchange | 8 strings → 4 | 2 active + 2 meso (pseudoasymmetric) |

Table 6.15 — one method, four different groups.

The last row is the subtle one. For an odd-n chain the exchange operation fixes the palindromic strings, but the middle carbon of a palindrome is pseudoasymmetric, so the two palindromes (R,S,R) and (S,R,S) are diastereomers, not the same compound. The orbit count therefore gives 8 arrangements → 6 orbits, matching active 6 + meso 2 in the Part I table when the ends are similar; the arithmetic is set out on sheet J-19.

KEY RESULT
Orbit counting is not "divide by 2". Each symmetry operation has its own fixed-point count, and the average of those counts is the number of distinct objects. For a chain that average is $(2^n + F)/2$.

## 6.30 Faces and groups

A carbonyl carbon is planar; its two faces are **enantiotopic** if the molecule has no other stereocentre, because attack on one face gives one enantiomer and attack on the other gives the other. The faces are labelled Re and Si. An enzyme can distinguish them, which is why biological reductions are stereospecific.

Two identical groups on a tetrahedral carbon are **enantiotopic** if replacing one or the other by a different group gives enantiomers, and **diastereotopic** if it gives diastereomers. In ethanol the two H on $CH_2$ are enantiotopic; in (R)-butan-2-ol the two H on C1 are diastereotopic, which is why they show different NMR signals.

| Relationship | Replacement gives | NMR: same signal? |
|---|---|---|
| homotopic | the same compound | yes |
| enantiotopic | enantiomers | yes in an achiral solvent |
| diastereotopic | diastereomers | no — different chemical shifts |

Table 6.16 — the three relationships between "identical" groups.

## 6.31 Why this matters for mechanism

Every stereospecific enzyme reaction is a statement about enantiotopic groups: the enzyme recognises one of two otherwise identical groups or faces. In the laboratory, an achiral reagent cannot distinguish enantiotopic faces, so it gives a racemic mixture — which is exactly why asymmetric catalysis is hard and why it won a Nobel prize.

## 6.32 Pseudoasymmetric centres

A carbon attached to two groups that are themselves chiral, and mirror images of each other (one R, one S), plus two other different groups, is **pseudoasymmetric**. It is labelled r or s (lower case) and it is not a stereocentre in the usual sense: the molecule containing it is achiral.

The standard example is meso-tartaric acid: the central C2-C3 bond has (R) on one side and (S) on the other, and the two carbons are pseudoasymmetric. This is why the meso form of a three-centre chain can exist in two diastereomeric forms (R,r,S) and (R,s,S) — a subtlety JEE Advanced has used more than once.

KEY RESULT
Pseudoasymmetric centre: two ligands are enantiomorphic (R and S), the other two differ. The centre is labelled r/s and the molecule stays achiral. It is what makes the odd-n meso row of the counting table read $2^{((n-1)/2)}$ rather than 1.

TRAP
An r/s centre is not a stereocentre and does not make the molecule chiral. If a question asks "how many stereocentres?", do not count it; if it asks "how many stereoisomers?", it is exactly the subtlety that decides the answer.

## 6.33 Descriptors of purity

| Quantity | Definition | Range | Use |
|---|---|---|---|
| optical purity | observed [α] / [α] of pure enantiomer | 0–100 % | historical; equals ee for two-component mixtures |
| enantiomeric excess (ee) | ([R] − [S]) / ([R] + [S]) | 0–100 % | modern standard |
| diastereomeric excess (de) | ([major] − [minor]) / total | 0–100 % | for reactions creating a second centre |
| enantiomeric ratio (er) | [R] : [S] | 1:1 to ∞ | used in catalysis papers |

Table 6.17 — four numbers, one idea: how far from racemic.

The relation between ee and the observed rotation is linear: ee = $\alpha_{obs}/\alpha_{pure}$. That is why polarimetry was the standard method for a century, and why it fails when an achiral impurity changes the concentration or when the sample contains a meso form.

## 6.34 Kinetic resolution

If the two enantiomers react at different rates with a chiral reagent or enzyme, stopping the reaction at partial conversion leaves the slower enantiomer enriched and converts the faster one into a product. The enantiomeric excess of the recovered starting material rises with conversion; at 50 % conversion the theoretical maximum is 100 % of the slower enantiomer, but the yield is only 50 %.

ee of recovered substrate rises as conversion increases
ee of product falls as conversion increases

Dynamic kinetic resolution combines racemisation of the substrate with selective reaction, so both enantiomers are funnelled into one product and the yield can exceed 50 %.

## 6.35 Chiroptical methods

Optical rotatory dispersion (ORD) plots [α] against wavelength; circular dichroism (CD) measures the different absorption of left- and right-circularly polarised light. Both give a spectrum, not a single number, and both can identify a chromophore's handedness — which is how the absolute configuration of complex natural products is assigned. They are outside the JEE syllabus but explain why "specific rotation" is only the simplest chiroptical measurement.

KEY RESULT
ee is the number, [α] is the measurement, and ORD/CD are the spectra. For JEE, ee = $\alpha_{obs}/\alpha_{pure}$ and the sign of α tells you which enantiomer is in excess — not the configuration.

## 6.36 Problems

1. How many stereoisomers does 1,2-dimethylcyclopropane have, and which are optically active?
2. Assign R/S to both centres of meso-tartaric acid and explain why the molecule is achiral.
3. trans-2-Butene reacts with $Br_2$. Give the product, its stereochemistry, and whether the product mixture is optically active.
4. How many stereoisomers does 1,2,3-tribromocyclopropane have?
5. A sample of 2-butanol shows $[\alpha] = +3.4°$, and the pure enantiomer has $[\alpha] = +13.5°$. What is the ee and the composition?
6. Why is trans-cyclooctene chiral while cis-cyclooctene is not?
7. $S_N1$ hydrolysis of (R)-2-bromobutane gives a product of $[\alpha] = -2.0°$ while pure (S)-2-butanol has $[\alpha] = -13.5°$. What is the ee, and what does it tell you about the mechanism?
8. How many stereoisomers does 2,3,4-trihydroxypentanedioic acid have?

## 6.37 Solutions

**(1)** Three: cis (achiral, a plane through C3 and the C1-C2 bond) and a trans pair (chiral). Active: the trans pair, 2 of the 3.

**(2)** (2R,3S). Each centre has OH > COOH > H; reading the Fischer with H horizontal reverses the apparent direction. The molecule has a plane through the middle of the C2-C3 bond, so the two halves are mirror images and the mirror image of the whole is itself.

**(3)** Anti addition gives the meso (2R,3S) dibromide — one compound, achiral, so the product is optically inactive (and not because it is racemic: it is a single achiral compound).

**(4)** Three geometrical arrangements (all-cis; one trans; the 1,2-trans-3-cis pattern) collapse to four stereoisomers: all-cis is achiral (plane), the other two arrangements are chiral pairs except where a plane survives. Draw each and check: the answer is 4, of which 2 are active.

**(5)** ee = 3.4/13.5 = 25 %; the sample is 62.5 % of the (+) enantiomer and 37.5 % of the (−) enantiomer.

**(6)** The trans double bond forces the ring above and below the alkene plane, so the molecule has no mirror plane and is chiral; the cis alkene leaves a plane through the ring and the double bond, so cis-cyclooctene is achiral.

**(7)** ee = 2.0/13.5 = 15 %. The product is mostly the (S) enantiomer, which is the inverted product, but only 15 % ee means the reaction is close to racemic — consistent with $S_N1$ through a planar carbocation, with a small excess of inversion because the leaving group shields one face briefly.

**(8)** $n = 3$ with similar ends: active $2^2 - 2^1 = 2$, meso $2^1 = 2$, total 4. The active pair is (R,R,R)/(S,S,S); the two meso forms have an (R) and an (S) end and a pseudoasymmetric middle carbon, so (R,r,S) and (R,s,S) are diastereomers, not the same compound.

KEY RESULT
Every solution above reduces to one of four moves: count centres, test symmetry, assign CIP, or convert a rotation into an ee. If a solution needs a fifth move, the question is testing a definition you have not stated.

## 6.38 Organic synthesis

Every asymmetric synthesis is an answer to the question "how do you make one enantiomer?". The strategies are: use a chiral starting material from the chiral pool (amino acids, sugars, terpenes); use a chiral auxiliary that is removed afterwards; use a chiral catalyst (organocatalyst or transition-metal complex); or resolve at the end. The first three are stereoselective, the fourth is a separation.

The contraceptive pill, statins, and most modern drugs are single enantiomers because the wrong enantiomer can be inactive or harmful. That commercial fact is why asymmetric catalysis is a large industry, and why the vocabulary of this chapter (ee, de, enantioselective) appears in every process chemistry paper.

## 6.39 Polymers

The stereochemistry of a polymer chain decides whether it crystallises. Isotactic polypropene (all methyls on one side, made with a Ziegler–Natta catalyst) is crystalline and rigid; atactic polypropene (random) is amorphous and rubbery. The monomer is achiral, but the polymer is not — tacticity is stereochemistry at the level of a chain.

## 6.40 Biochemistry

Life is homochiral: proteins are made of L-amino acids and nucleic acids of D-ribose. An enzyme's active site is itself chiral, so it recognises one enantiomer of a substrate — which is why (*R*)-thalidomide is sedating and (*S*)-thalidomide is teratogenic, and why drug regulators now require single-enantiomer data.

## 6.41 Materials and photochemistry

Chiral liquid crystals, chiral dopants for displays, and photoresponsive switches all use the same ideas. Azobenzene is the standard switch: trans is flat and mesogenic, cis is bent and disrupts the phase; irradiation at one wavelength switches the shape, and the thermal back-reaction restores it.

## 6.42 Inorganic chemistry

Tris-chelate complexes $[M(en)_3]^{n+}$ are chiral ($\Lambda/\Delta$), as are many octahedral complexes with bidentate ligands. Coordination chemistry also supplies examples of stereochemistry without carbon (sulfoxides, phosphines), and of fluxional molecules whose stereochemistry averages on the NMR timescale.

> KEY RESULT
> Stereochemistry is not a chapter of organic chemistry; it is a way of describing shape that reappears in polymers, enzymes, materials and coordination compounds. The vocabulary is the transferable part.

## 6.43 The five-question audit

1. **Have I drawn the molecule?** Every error in this chapter starts with an unexamined drawing.
2. **Have I tested symmetry before counting?** A plane or a centre removes half the count.
3. **Have I asked whether the ends are identical?** It changes the counting row and the existence of meso forms.
4. **Have I checked whether the barrier makes the arrangement configurational?** Atropisomers and amides are configurations; ethane conformers are not.
5. **Have I stated the mechanism before the stereochemical outcome?** Inversion, racemisation, syn and anti are consequences of mechanism, not facts to recall.

## 6.44 Extension-level traps

- A twist-boat is chiral but a cyclohexane sample is not — the chiral conformation is not long-lived.
- Two enantiomers of a biphenyl are atropisomers only if the barrier is above ~80 kJ $mol^{-1}$.
- trans-Cyclooctene is chiral; cis-cyclooctene is not. The double bond geometry, not the ring size, does the work.
- Pseudoasymmetric centres (r/s) are not stereocentres but they change the stereoisomer count.
- Burnside counting needs the whole symmetry group, not just the end exchange; for a ring the rotations matter.
- The anomeric effect can override a steric preference; do not quote A-values alone for a pyranose.
- Stereospecific and stereoselective are not synonyms; examiners award marks for the distinction.

## 6.45 What remains uncertain

Three things in this set are worth flagging honestly. First, numerical energy values (barriers, A-values, strain energies) are quoted from standard textbook ranges; different texts differ in the second digit, so use them to fix order of magnitude, not to settle a tie. Second, the syn/anti convention for oximes is not uniform across Indian textbooks, and the notes therefore teach E/Z as the primary descriptor. Third, the CIP rules for axes and for pseudoasymmetric centres are stated here in outline only; the full IUPAC blue-book rules include several tie-breaking refinements that are beyond the JEE syllabus.

> KEY RESULT
> A full-mark answer names the idea, shows the drawing, and states the conclusion in the vocabulary the question used. Everything else is detail.

This is the end of Part II. Part I (J-01 to J-29) covers the JEE Main and JEE Advanced core; the source map, the corrections made and the verification status are in SCOPE.md and SOURCE-AUDIT.md, and the printable PNGs and PDFs are indexed in index.html.